DIFFERENTIABILITY

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Lecture Outline

1. Introduction
2. Derivatives of elementary functions
3. Differentiation Rules
4. Implicit Differentiation
5. Higher-Order derivatives



Outline of Presentation

1 Introduction

2 Derivatives of elementary functions

3 Differentiation Rules

4 Implicit Differentiation

5 Higher-Order derivatives



Introduction

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- Suppose the relation is expressed as a function then the derivative describes the rate of change of the dependent variable relative to the independent variable.
- Pictorially, this is the slope/ tangent to the graph of the function at given points.
- Recall that $\text{slope} = \dfrac{\triangle y}{\triangle x} = \dfrac{y_2 - y_1}{x_2 - x_1}$
- The slope of the line passing through the point $(x_0, f(x_0))$ and touching the graph of f is

$$\text{Slope} = \frac{f(x) - f(x_0)}{x - x_0}$$



Introduction

Definition

Let $f(x)$ be defined at $x_0 \in (a, b)$. Then the derivative of $f(x)$ at x_0 denoted by $f'(x_0)$ is defined by

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

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Suppose x close to x_0 by some small value h, then $x = x_0 + h$ and so we have

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h} \tag{1}$$

provided the limit exist.



Introduction

Remarks

1. $f'(x_0) = \lim\limits_{x \to x_0} \dfrac{f(x) - f(x_0)}{x - x_0}$ is the derivative of f at x_0.
2. $f'(x_0^-) = \lim\limits_{x \to x_0^-} \dfrac{f(x) - f(x_0)}{x - x_0}$ is the **left-derivative** of f at x_0.
3. $f'(x_0^+) = \lim\limits_{x \to x_0^+} \dfrac{f(x) - f(x_0)}{x - x_0}$ is the **right-derivative** of f at x_0.



Introduction

Definition

1. Differentiation is the process of finding the derivative of a function.
2. f is said to be differentiable at x_0 when $f'(x_0)$ exists.
3. A function $f(x)$ is differentiable in an interval (a, b) if it is differentiable at every point in (a, b).



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Notations

The following notations can be used for derivative of $f(x)$

$$1)\ f'(x), \qquad 2)\ \frac{df(x)}{dx}, \qquad 3)\ D_x f(x), \qquad 4)\ Df(x)$$

Example

Find the derivative of $f(x) = x^2 + 3$ at $x = 2$.



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We need to determine $f'(2) = \lim_{h \to 0} \frac{f(2+h) - f(2)}{h}$. We have

$$f'(x_0) = \lim_{h \to 0} \frac{f(2+h) - f(2)}{h} \tag{2}$$

$$= \lim_{h \to 0} \frac{h^2 + 4h + 4 - 4}{h} \tag{3}$$

$$= \lim_{h \to 0} (h + 4) = 4 \tag{4}$$



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The derivative of a function f at $(x_0, f(x_0))$ is the slope of line through $(x_0, f(x_0))$.



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$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \to 0} \frac{\sqrt{x+h-1} - \sqrt{x-1}}{h}$$



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&= \lim_{h\to 0} \frac{x+h-1-(x-1)}{h\left(\sqrt{x+h-1}+\sqrt{x-1}\right)} \\
&= \frac{1}{\sqrt{x-1}+\sqrt{x-1}} = \frac{1}{2\sqrt{x-1}}
\end{aligned}
$$



Thus $f'(x)$ is defined if $x > 1$. Therefore the function $f(x) = \sqrt{x-1}$ is differentiable in $(1, \infty)$. This is smaller than the domain of f which is $[1, \infty)$.



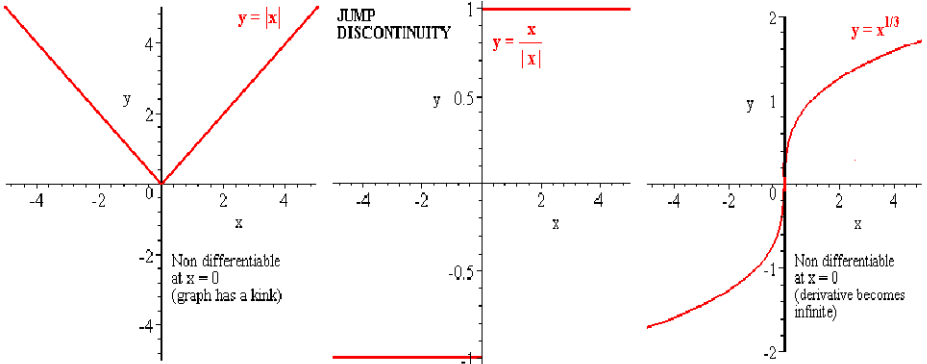
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Thus a function is not differentiable at a point if

1. the graph has **corner** or a **kink**
2. the function is discontinuous there
3. there is a vertical tangent line to the function at that point



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Differentiation of constant functions

The derivatives of constants function $f(x) = c$ are given by

$$\frac{d}{dx} f(x) = 0$$

for c constant.

that is

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \frac{c-c}{h} = 0 \tag{5}$$



Differentiation of Power functions

Power functions are functions of the form

$$f(x) = x^n \tag{6}$$

where $n \in \mathbb{R}$. Then we have

$$\frac{d}{dx} f(x) = \frac{d}{dx} x^n \tag{7}$$

$$= nx^{n-1} \tag{8}$$



Example

1) $f(x) = x^4$, 2) $f(x) = \dfrac{1}{x^5}$, 3) $f(x) = \sqrt[3]{x}$

1. $f(x) = x^4 \Longrightarrow \dfrac{d}{dx} f(x) = 4x^{4-1} = 4x^3.$



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2. $f(x) = x^{-5} \Longrightarrow \dfrac{d}{dx} f(x) = -5x^{-5-1} = -\dfrac{5}{x^6}.$



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2. $f(x) = x^{-5} \implies \dfrac{d}{dx} f(x) = -5x^{-5-1} = -\dfrac{5}{x^6}.$
3. $f(x) = x^{1/3} \implies \dfrac{d}{dx} f(x) = \dfrac{1}{3} x^{(1/3)-1} = \dfrac{1}{3} x^{-2/3} = \dfrac{1}{3\sqrt[3]{x^2}}.$



Differentiation of Trigonometric functions

Below are the derivatives of some trigonometric functions

1. $\dfrac{d}{dx}\sin x = \cos x$
2. $\dfrac{d}{dx}\cos x = -\sin x$
3. $\dfrac{d}{dx}\tan x = \sec^2 x$



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4. $\frac{d}{dx}\csc x = -\csc x \cot x$
5. $\frac{d}{dx}\sec x = \sec x \tan x$
6. $\frac{d}{dx}\cot x = -\csc^2 x$



Differentiation of Inverse Trigonometric Functions

1. $\dfrac{d}{dx}(\sin^{-1} x) = \dfrac{1}{\sqrt{1-x^2}}$
2. $\dfrac{d}{dx}(\cos^{-1} x) = -\dfrac{1}{\sqrt{1-x^2}}$
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4. $\dfrac{d}{dx}(\sec^{-1} x) = \dfrac{1}{|x|\sqrt{1-x^2}}$
5. $\dfrac{d}{dx}(\csc^{-1} x) = -\dfrac{1}{|x|\sqrt{1-x^2}}$
6. $\dfrac{d}{dx}(\cot^{-1} x) = -\dfrac{1}{1+x^2}$



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3. $$\frac{d}{dx}\ln x = \frac{\frac{d}{dx}(x)}{x} = \frac{1}{x} \tag{11}$$



Differentiation of Hyperbolic Functions

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5. $\frac{d}{dx}(\operatorname{csch} x) = -\operatorname{csch} x \coth x$
6. $\frac{d}{dx}(\coth x) = -\operatorname{csch}^2 x$



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3. $\dfrac{d}{dx}(\tanh^{-1} x) = \dfrac{1}{1-x^2}; \quad |x| < 1$
4. $\dfrac{d}{dx}(\operatorname{sech}^{-1} x) = -\dfrac{1}{x\sqrt{1-x^2}}; \quad 0 < x < 1$
5. $\dfrac{d}{dx}(\operatorname{csch}^{-1} x) = -\dfrac{1}{|x|\sqrt{1-x^2}}, \quad x \neq 0$
6. $\dfrac{d}{dx}(\coth^{-1} x) = \dfrac{1}{1-x^2}; \quad |x| > 1$



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Differentiation of sums of functions

For functions f and g, and constants α, β

$$\frac{d}{dx}(\alpha f \pm \beta g) = \alpha \frac{d}{dx} f \pm \beta \frac{d}{dx} g \tag{12}$$



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We have that

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$$\begin{aligned} f'(x) &= \frac{d}{dx}x^4 + 6\frac{d}{dx}\sin x + \frac{d}{dx}4 - \frac{d}{dx}\ln x \\ &= 4x^3 + 6\cos x + 0 - \frac{1}{x} = 4x^3 + 6\cos x - \frac{1}{x} \end{aligned}$$



Differentiation of products of functions

The derivative of product of functions f and g is given by

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We notice that $h(x)$ is the product of the functions $f(x) = \sin x$ and $g(x) = \cos x$. Hence applying the product rule we have

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$$\begin{aligned}\frac{d}{dx}h(x) &= f\frac{d}{dx}g + g\frac{d}{dx}f \\ &= (e^x + x^2)(3^x \ln 3 + \sec^2 x) + (3^x + \tan x)(2x + e^x)\end{aligned}$$



Differentiation of Rational functions

The derivative of rational functions is given by

$$\frac{d}{dx}\left(\frac{f}{g}\right) = \frac{g\frac{d}{dx}f - f\frac{d}{dx}g}{g^2} \tag{14}$$



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Find $h'(x)$ if $h(x) = \tan x$

Recall that $h(x) = \dfrac{\sin x}{\cos x}$. Hence applying the quotient rule we have



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Example

Find $h'(x)$ if $h(x) = \dfrac{x^3 + x + 2}{x^2 + 4}$



Example

Find $h'(x)$ if $h(x) = \dfrac{x^3 + x + 2}{x^2 + 4}$

Notice that $h(x)$ is the quotient of the functions $f(x) = x^3 + x + 2$ and $g(x) = x^2 + 4$. Hence applying the quotient rule we have

$$\begin{aligned} \frac{d}{dx}h(x) &= \frac{g\dfrac{d}{dx}f - f\dfrac{d}{dx}g}{g^2} \\ &= \frac{(x^2+4)(3x^2+1) - (x^3+x+2)(2x)}{(x^2+4)^2} \end{aligned}$$



Differentiation of composite functions

1. Given two function $f(x)$ and $u(x)$, then their composite function defined by

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Note

$h(x)$ is not the product of functions. The functions f and u are not multiplying but rather u is "placed" in f as an argument.

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Find $h'(x)$ if $h(x) = e^{\cos x}$



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Derivatives of inverse functions

Suppose $y = f^{-1}(x)$ then $x = f(y)$. Thus

$$\frac{dy}{dx} = \frac{1}{dx/dy} = \frac{1}{f'(y)} \tag{17}$$



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1. $y = \ln x$
2. $y = \sin^{-1} x$
3. $y = \cos^{-1} x$



$$y = \ln x \implies x = e^y \tag{18}$$



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2 Derivatives of elementary functions

3 Differentiation Rules

4 Implicit Differentiation

5 Higher-Order derivatives



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Example

The equation $xy + e^y = 1$ **implicitly** defines the function y. There is no explicit formula to define y. But the derivative with respect to x can be still obtained as follows

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$$\frac{dy}{dx} = -\frac{y}{x + e^y} \tag{35}$$

$$= -\frac{y}{x + 1 - xy} \tag{36}$$



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Higher-Order derivatives

Notice that the derivatives of functions are also functions. Hence we can find the derivatives of the derivatives of functions. These are precisely the higher derivatives of the given function.

Second order derivatives can be denoted by any of the following

1) $f''(x)$, 2) $\frac{d^2 f(x)}{dx^2}$, 3) $D_x^2 f(x)$, 4) $D^2 f(x)$

Third order derivatives can be denoted by any of the following

1) $f'''(x)$, 2) $\frac{d^3 f(x)}{dx^3}$, 3) $D_x^3 f(x)$, 4) $D^3 f(x)$

n_{th} order derivatives can be denoted by any of the following

1) $f^n(x)$, 2) $\frac{d^n f(x)}{dx^n}$, 3) $D_x^n f(x)$, 4) $D^n f(x)$

Example

$$y = x^3 - 6x^2 + 9x \tag{37}$$



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$$y = x^3 - 6x^2 + 9x \tag{37}$$

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$$\frac{d^2y}{dx^2} = 3x - 12 \tag{39}$$

$$\frac{d^3y}{dx^3} = 3 \tag{40}$$

$$\frac{d^ny}{dx^n} = 0; \quad n \geq 4 \tag{41}$$



Example

$$y = \sin x \tag{42}$$

$$\frac{dy}{dx} = \cos x \tag{43}$$

$$\frac{d^2y}{dx^2} = -\sin x \tag{44}$$

$$\frac{d^3y}{dx^3} = -\cos x \tag{45}$$

$$\frac{d^4y}{dx^4} = \sin x \tag{46}$$

$$\frac{d^5y}{dx^5} = \cos x \tag{47}$$



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$$\frac{d^4y}{dx^4} = \sin x \tag{46}$$

$$\frac{d^5y}{dx^5} = \cos x \tag{47}$$

You notice that the higher derivatives does not vanish follows some pattern, that is

1. $f^n(x) = \sin x \quad n = 4k$ for some k
2. $f^n(x) = \cos x \quad n = 4k+1$ for some k
3. $f^n(x) = -\sin x \quad n = 4k+2$ for some k
4. $f^n(x) = -\cos x \quad n = 4k+3$ for some k



Suppose $y(x)$ is a product of the functions f and g that is $y = fg$. Then

$$Dy = gDf + fDg \tag{48}$$



Suppose $y(x)$ is a product of the functions f and g that is $y = fg$. Then

$$\begin{aligned} Dy &= gDf + fDg \\ D^2y &= D((gDf + f_Dg)) \end{aligned} \tag{48}$$



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Suppose $y(x)$ is a product of the functions f and g that is $y = fg$. Then

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\begin{aligned}
Dy &= gDf + fDg && (48)\\
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D^3y &= D(gD^2f + 2DfDg + fD^2g) = gD^3f + 3D^2fDg + 3DfD^2g + fD^3g && (50)\\
\vdots &= \vdots\\
D^ny &= gD^nf + \binom{n}{1}DgD^{n-1}f + \cdots + fD^ng && (51)
\end{aligned}
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Theorem (Leibnitz theorem)

Let $y = fg$ where f and g are functions of x, then the nth derivative of y is given by

$$D^ny = \sum_{r=0}^{n}\binom{n}{r}D^rgD^{n-r}f$$

Example

Find the third derivative of $y = x^2 e^x$



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$$= e^x (x^2 + 6x + 6) \tag{55}$$



Exercise

1. Find the first derivatives of 1) $f(x)=\dfrac{x^2+x-2}{x^3+6}$ 2) $f(x)=\dfrac{e^x}{1+x^3}$

   3) $f(x)=\left(\dfrac{1}{x^2+1}\right)(\tan x)$

2. Find the implicit derivative of the function y or θ with respect to x or r :

   1) $y^2-x^2=\sin(xy)$ 2) $x^3+y^3-9xy=0$ 3) $x^2y+y^2x=6$

   4) $x+\tan(xy)=0$ 5) $x\cos(2x+3y)=y\sin x$ 6) $\cos r+\cot\theta=r.$

3. Find the general formula for the nth derivatives of the following.
   1. $y=\ln x$
   2. $y=e^{ax}$
   3. $y=\sin x$



END OF LECTURE
THANK YOU

